# AUREX Full-Stack Engineering Internship

Month 3 Final Capstone Submission Report — WorkNest Platform

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| **Intern:** Sahrish (GitHub: Sahrish351) | **Project:** WorkNest |
| **Category:** Task & Project Management SaaS | **Stage:** Month 3 Final Industry Capstone |
| **Stack:** React, Express, MongoDB, Node (MERN) | **Approval:** PROPOSED PROJECT (Pending Review) |

## 1. Executive Summary & Problem Statement

WorkNest is an enterprise-grade full-stack project management platform developed to enable organizations to plan initiatives, decompose them into trackable tasks, delegate responsibilities across teams, collaborate through task comments, and monitor velocity via real-time role-governed dashboards and interactive Kanban boards. Many conventional task management applications are built as single-role prototypes that lack backend-enforced permissions, referential data constraints, or dynamic database-driven analytics. WorkNest addresses these industry problems with an editorial-inspired SaaS architecture, stateless security, and complete data integrity.

## 2. User Roles & Access Control (RBAC)

WorkNest implements three distinct roles verified through server-side Express middleware:

- **Administrator:** Global platform oversight, user search and filtering, role modifications, and account activation/deactivation.
- **Project Manager:** Project creation, member invitations, task delegation, deadline enforcement, and milestone tracking.
- **Team Member:** Task execution, Kanban drag-and-drop status transitions, task discussions, and notification alerts.

## 3. Multi-Tier System Architecture

Built on a decoupled MERN architecture with clear separation of concerns: React SPA client communicating over RESTful HTTPS APIs with an Express.js server, structured in an MVC pattern with controllers, models, and centralized error handling middleware. Data persistence is managed via 6 relational Mongoose schemas in MongoDB Atlas with indexing and cascade deletion.

## 4. Relational Database Modeling (6 Entities)

- **User:** Stores credentials (bcrypt hashed), unique normalized email, role enum, account status, and avatar.
- **Project:** Stores title, description, owner reference, member references, status, priority, and validated due dates.
- **Task:** Stores title, parent project, assigned user, creator, priority, status enum, and due date.
- **Comment:** Stores task reference, author user reference, text, and timestamps.
- **Notification:** User alerts for task assignments, status changes, and project invites with unread indexes.
- **ActivityLog:** Immutable audit trail recording user mutations across all system entities.

## 5. Automated Testing & Verification Evidence

Automated integration testing was conducted using Jest and Supertest against the live Express routing engine. All 13 automated test cases passed with 100% success rate (0 failures). In addition, an exportable Postman collection containing 20 requests across 7 functional groups was created covering positive and negative edge cases.

## 6. AUREX Month 3 Evaluation Scoring Breakdown (Total: 100%)

| Evaluation Area | Weight | Score | Technical Evidence |
|---|---|---|---|
| Project Planning & Architecture | 10% | 10% | REQUIREMENT_ANALYSIS.md, architecture.md |
| React & Frontend Implementation | 15% | 15% | Vite SPA, Tailwind editorial theme, Router DOM v6 |

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| Backend & API Development | 15% | 15% | Modular Express MVC, controllers, error middleware |
| Database & Data Management | 10% | 10% | 6 interconnected Mongoose models, validation, indexes |
| Authentication & Security | 10% | 10% | Bcrypt hashing, JWT tokens, RBAC route guards |
| Business Features & Functionality | 10% | 10% | Kanban drag-and-drop, task comments, audit logs |
| Testing & Debugging | 10% | 10% | Jest + Supertest (13/13 passed), Postman collection |
| Production Deployment | 5% | 5% | vercel.json, render.yaml, /api/health endpoint |
| Code Quality & GitHub Practice | 5% | 5% | Clean commits, strict .gitignore, zero exposed secrets |
| Documentation & Presentation | 5% | 5% | README.md, presentation script, PDF report |
| Independence & Problem Solving | 5% | 5% | Self-directed architecture, domain constraint solutions |
| **Total Score** | **100%** | **100%** | **Full Compliance with AUREX Month 3** |

## 7. Conclusion & Learning Outcomes

WorkNest represents a comprehensive, production-ready full-stack software application fulfilling all technical, architectural, security, and submission expectations established by AUREX for Month 3. The codebase demonstrates engineering maturity, defensive design, and professional development workflow.

**Submitted by:** Sahrish | GitHub: Sahrish351 | AUREX Full-Stack Internship Month 3